

Docker - Create an image

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ mkdir my_docker_image
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ cd my_docker_image
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$ nano Dockerfile
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$
```

Create [Dockerfile.py](#)

Create [server.py](#) file

In subfolder containing Dockerfile, create the image:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$ docker image build . -t my_image:latest
[+] Building 7.0s (9/9) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 215B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/4] FROM docker.io/library/python:3.9-slim@sha256:a40cf9eba2c3ed9226afa9ace504f07ad30fe831343bb1c69f7a6707aad7c21
=> [internal] load build context
=> => transferring context: 31B
=> CACHED [2/4] WORKDIR /my_server/
=> CACHED [3/4] COPY server.py .
=> [4/4] RUN pip install flask==2.0.0 werkzeug==2.0.3
=> exporting to image
=> => exporting layers
=> => writing image sha256:c3603c15ed589826b79bb903fdf301033172fae32027331ee6aa473bf5894381
=> => naming to docker.io/library/my_image:latest
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$ docker image ls
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
my_image             latest          c3603c15ed58    33 seconds ago  137MB
<none>              <none>         d173878dfc8a    5 minutes ago   137MB
dpape/pgadmin4      latest          46af92358e2e    2 months ago    524MB
postgres            16-alpine       2875f9e036c2    3 months ago    275MB
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker/my_docker_image$ docker run -p 5000:5000 my_image:latest
* Serving Flask app 'my_server' (lazy loading)
* Environment: production
  WARNING: This is a development server. Do not use it in a production deployment.
  Use a production WSGI server instead.
* Debug mode: off
* Running on all addresses.
  WARNING: This is a development server. Do not use it in a production deployment.
* Running on http://172.17.0.2:5000/ (Press CTRL+C to quit)
```

In new terminal run:

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ curl http://localhost:5000
Hello World from a containerized server
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image save --output my_docker_image.tar my_image
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image rm -f my_image
Untagged: my_image:latest
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker image load --input my_docker_image.tar
Loaded image: my_image:latest
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```

```
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker login
Log in with your Docker ID or email address to push and pull images from Docker Hub. If you don't have a Docker ID, head over to https://hub.docker.com/ to create one.
You can log in with your password or a Personal Access Token (PAT). Using a limited-scope PAT grants better security and is required for organizations using SSO. Learn more at https://docs.docker.com/go/access-tokens/
```

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Login Succeeded
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker tag my_image [redacted]/my_image:latest
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$ docker push [redacted]/my_image:latest
The push refers to repository [docker.io/[redacted]/my_image]
7b7f4bbae8d2: Pushed
fc3894f707b6: Pushed
51bf60077405: Pushed
f0d3b09df3b4: Mounted from library/python
b9a50396677d: Mounted from library/python
be76169c9d2c: Mounted from library/python
7fb72a7d1a8e: Mounted from library/python
latest: digest: sha256:0f7734b5f57902a9fd3d463ddfaf46409fc4407f296da4a01a7927e33eedefed size: 1783
ubuntu@ip-172-31-18-113:~/datascientest-data-engineering/docker$
```